

Ex. No. 5 — Calculation of Group Velocity, Phase Velocity and Phase Constant of EM Waves in Rectangular Waveguides

Test Case 1

A rectangular waveguide has dimensions $a = 4.0$ cm, $b = 2.0$ cm and operates at $f = 8$ GHz. (1) Calculate the cutoff frequency for the TE_{10} mode. (2) Predict the phase and group velocities for TE_{10} at this frequency. (3) Examine the effect of reducing the width to $a = 3.0$ cm on the cutoff frequency and propagation characteristics.

Scilab Program

```
clc; clear; close;
a = 4.0*10^-2; b = 2.0*10^-2; f = 8*10^9; // meters, meters, Hz
V = 3*10^8;
m = input("enter the value of m= "); n = input("enter the value of n= ");
fc = (V/2)*sqrt((m/a)^2+(n/b)^2);
disp("Cut off frequency (Hz) =", fc);
A = V/sqrt(1-(fc/f)^2); // phase velocity
disp("Phase velocity (m/s) =", A);
B = V*V/A; // group velocity
disp("Group velocity (m/s) =", B);
```

Case 1 — Original Dimensions ($a = 4.0$ cm, $b = 2.0$ cm), $m = 1$, $n = 0$, $f = 8$ GHz

Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f_c)	3.7500 GHz	3.7500 GHz
Phase Velocity (v_p)	339.6236 Mm/s	339.6236 Mm/s
Group Velocity (v_g)	264.9993 Mm/s	264.9993 Mm/s

Case 2 — Reduced Width ($a = 3.0$ cm, $b = 2.0$ cm), $m = 1$, $n = 0$, $f = 8$ GHz

Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f_c)	5.0000 GHz	5.0000 GHz
Phase Velocity (v_p)	384.3076 Mm/s	384.3076 Mm/s
Group Velocity (v_g)	234.1874 Mm/s	234.1874 Mm/s

Variation of Cutoff Frequency, Phase and Group Velocity with Width (a) ($b = 2.0$ cm, $f = 8$ GHz, TE_{10})

Width (a)	Sim. f_c (GHz)	Sim. v_p (Mm/s)	Sim. v_g (Mm/s)	Theo. f_c (GHz)	Theo. v_p (Mm/s)	Theo. v_g (Mm/s)
4.0 cm	3.7500	339.6236	264.9993	3.7500	339.6236	264.9993
3.5 cm	4.2857	355.2821	253.3198	4.2857	355.2821	253.3198
3.0 cm	5.0000	384.3076	234.1874	5.0000	384.3076	234.1874
2.5 cm	6.0000	453.5574	198.4313	6.0000	453.5574	198.4313
2.0 cm	7.5000	862.1054	104.3956	7.5000	862.1054	104.3956

Test Case 2

A radar system operates at $f = 15$ GHz with waveguide dimensions $a = 2.5$ cm, $b = 1.25$ cm. (1) Compute the cutoff frequency for the TE_{10} mode. (2) Determine the phase and group velocities for TE_{10} . (3) Discuss strategies to maintain signal fidelity when dispersion is critical to radar accuracy.

Scilab Program

```
clc; clear; close;
a = 2.5*10^-2; b = 1.25*10^-2; f = 15*10^9; // meters, meters, Hz
V = 3*10^8;
m = input("enter the value of m= "); n = input("enter the value of n= ");
fc = (V/2)*sqrt((m/a)^2+(n/b)^2);
disp("Cut off frequency (Hz) =", fc);
A = V/sqrt(1-(fc/f)^2); // phase velocity
disp("Phase velocity (m/s) =", A);
B = V*A; // group velocity
disp("Group velocity (m/s) =", B);
```

Output for TE_{10} Mode ($a = 2.5$ cm, $b = 1.25$ cm, $f = 15$ GHz)

Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f _c)	6.0000 GHz	6.0000 GHz
Phase Velocity (v _p)	327.3268 Mm/s	327.3268 Mm/s
Group Velocity (v _g)	274.9545 Mm/s	274.9545 Mm/s

Test Case 3

A waveguide with dimensions $a = 5.0$ cm, $b = 2.5$ cm carries a signal at $f = 10$ GHz. (1) Determine the number of TE modes that can propagate. (2) For TE_{10} , calculate the phase and group velocities. (3) Discuss how multiple propagating modes affect transmission quality and how to limit multimode effects.

Scilab Program

```
clc; clear; close;
a = 5.0*10^-2; b = 2.5*10^-2; f = 10*10^9; // meters, meters, Hz
V = 3*10^8;
m = input("enter the value of m= "); n = input("enter the value of n= ");
fc = (V/2)*sqrt((m/a)^2+(n/b)^2);
disp("Cut off frequency (Hz) =", fc);
A = V/sqrt(1-(fc/f)^2); // phase velocity
disp("Phase velocity (m/s) =", A);
B = V*V/A; // group velocity
disp("Group velocity (m/s) =", B);
```

Mode Propagation Table ($a = 5.0$ cm, $b = 2.5$ cm, $f = 10$ GHz)

Mode (m,n)	Sim. f_c (GHz)	Sim. Propagation	Theo. f_c (GHz)	Theo. Propagation
TE10	3.0000	Yes	3.0000	Yes
TE01	6.0000	Yes	6.0000	Yes
TE20	6.0000	Yes	6.0000	Yes
TE11	6.7082	Yes	6.7082	Yes
TE21	8.4853	Yes	8.4853	Yes
TE02	12.0000	No	12.0000	No

Result: Five TE modes can propagate at $f = 10$ GHz — TE10, TE01, TE20, TE11, TE21 (TE02 is cut off).

Output for TE_{10} Mode ($a = 5.0$ cm, $b = 2.5$ cm, $f = 10$ GHz)

Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f_c)	3.0000 GHz	3.0000 GHz
Phase Velocity (v_p)	314.4855 Mm/s	314.4855 Mm/s
Group Velocity (v_g)	286.1818 Mm/s	286.1818 Mm/s

Test Case 4

A research lab experiments with high-order TE modes ($m, n > 1$) in a waveguide with dimensions $a = 3.0$ cm, $b = 1.5$ cm, operating at $f = 12$ GHz. (1) Calculate the cutoff frequency for TE_{10} . (2) Compute its phase and group velocities. (3) Discuss how high-order modes can be used for mode multiplexing in communications.

Scilab Program

```
clc; clear; close;
a = 3.0*10^-2; b = 1.5*10^-2; f = 12*10^9; // meters, meters, Hz
V = 3*10^8;
m = input("enter the value of m= "); n = input("enter the value of n= ");
fc = (V/2)*sqrt((m/a)^2+(n/b)^2);
disp("Cut off frequency (Hz) =", fc);
A = V/sqrt(1-(fc/f)^2); // phase velocity
disp("Phase velocity (m/s) =", A);
B = V*A; // group velocity
disp("Group velocity (m/s) =", B);
```

Mode Propagation Table ($a = 3.0$ cm, $b = 1.5$ cm, $f = 12$ GHz)

Mode (m,n)	Sim. f_c (GHz)	Sim. Propagation	Theo. f_c (GHz)	Theo. Propagation
TE ₁₀	5.0000	Yes	5.0000	Yes
TE ₂₀	10.0000	Yes	10.0000	Yes
TE ₀₁	10.0000	Yes	10.0000	Yes
TE ₁₁	11.1803	Yes	11.1803	Yes
TE ₂₁	14.1421	No	14.1421	No
TE ₃₀	15.0000	No	15.0000	No

Output for TE_{10} Mode ($a = 3.0$ cm, $b = 1.5$ cm, $f = 12$ GHz)

Parameter	Simulation Value	Theoretical Value
Cutoff Frequency (f_c)	5.0000 GHz	5.0000 GHz
Phase Velocity (v_p)	330.0115 Mm/s	330.0115 Mm/s
Group Velocity (v_g)	272.7178 Mm/s	272.7178 Mm/s

Result

The cutoff frequency, phase velocity and group velocity of TE modes in rectangular waveguides were computed for varying waveguide dimensions and operating frequencies, and the number of propagating modes at each frequency was verified against the theoretical cutoff condition $f > f_c(m,n)$.